ECA1004 - Microprocessors and Microcontrollers

Vaniganji Sumanth Babu
19252447
Slot A

CO1 - Assessement Tool 3

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Viva Voice

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Q1. Data Transfer Group (e.g., MOV A, B)

Q2. ADD B (or ADD C / ADI 05H)

Q3. SUB B (or SUB C / SUI 05H)

Q4. CMA (1's complement / Bitwise NOT); Complement Accumulator A

Q5. ANA / ANI (AND operation); ANA B / ANI 0FH

Q6. ORA / ORI (OR operation); ORA B / ORI 80H

Q7. XOR operation (XRA / XRI); XRA B / XRI 0FH

Q8. JMP (Unconditional Jump); JMP 2050H

Q9. Conditional Branch instructions (e.g., JZ, JNZ); JZ 2050H, JNZ 2050H, JC 2050H

Q10. CMP / CPI; CMP B / CPI 20H
[P=1] 0C − 0C ⇒ Zero Flag (Z = 1)
[P=0] 0AH 0BH ⇒ AC = 1
[P=0] 0EH − 1DH ⇒ CY = 1, S = 1

Q11. Carry Flag (CY)

Q12. Immediate Addressing Mode (MVI A, 32H / ADI 05H)

Q13. Register Addressing Mode (MOV A, B / ADD C)

Q14. Direct Addressing Mode (LDA 2050H / STA 2050H)

Q15. Register Indirect Addressing Mode (MOV A, M where M → HL pair)

Q16. Rotate instructions (e.g., RLC, RRC)

Q17. XOR operation (XRA B / XRI data)

Q18. Branch / Program control Group (JMP 2050H, CALL 2050H, RET)

Q19. Masking using AND (ANA / ANI); ANA B / ANI 0FH

Q20. RET (Return), return from subroutine to address on stack.